File Edit View Git Project Build Debug Test Tools Extensions Window Help Search CalculatorLibrary Sign in

Debug Any CPU CalculatorLibrary

CalculatorTests.cs Test1.cs Calculator.cs

CalculatorLibrary CalculatorLibrary.Calculator Subtract(double a, double b)

```
using System;

namespace CalculatorLibrary
{
    2 references
    public class Calculator
    {
        2 references ✓ 2/2 passing
        public double Add(double a, double b) => a + b;
        2 references ✓ 2/2 passing
        public double Subtract(double a, double b) => a - b;
        1 reference ✓ 1/1 passing
        public double Multiply(double a, double b) => a * b;

        2 references ✓ 2/2 passing
        public double Divide(double a, double b)
        {
            if (b == 0)
                throw new DivideByZeroException("Cannot divide by zero.");
            return a / b;
        }
    }
}
```

100 % No issues found Ln: 19, Ch: 1 SPC CRLF UTF-8 with BOM

Output

Show output from: Build

```
Build started at 11:20 AM...
========== Build: 0 succeeded, 0 failed, 2 up-to-date, 0 skipped ==========
========== Build completed at 11:20 AM and took 00.045 seconds ==========
```

Solution Explorer

Search Solution Explorer (Ctrl+;)

Solution 'CalculatorLibrary' (2 of 2 proj

- CalculatorLibrary
  - Dependencies
  - Calculator.cs
- CalculatorLibrary.Tests
  - Dependencies
  - CalculatorTests.cs
  - MSTestSettings.cs
  - Test1.cs

Solution Explorer Git Changes

Properties

Calculator.cs File Properties

Advanced

| | |
|---|---|
| Build Action | C# compiler |
| Copy to Output Di | Do not copy |
| Custom Tool | |
| Custom Tool Name | |

Misc

| | |
|---|---|
| File Name | Calculator.cs |
| Full Path | C:\Users\Asus\source\re |

**Build Action**

How the file relates to the build and deployment processes.

Ready Test Explorer Add to Source Control Select Repository

File Edit View Git Project Build Debug Test Tools Extensions Window Help Search CalculatorLibrary Sign in

Debug Any CPU CalculatorLibrary

CalculatorTests.cs Test1.cs Calculator.cs

CalculatorLibrary.Tests CalculatorLibrary.Tests.CalculatorTests Setup()

```
using Microsoft.VisualStudio.TestTools.UnitTesting;
using CalculatorLibrary;
using System;

namespace CalculatorLibrary.Tests
{
    [TestClass]
    0 references
    public class CalculatorTests
    {
        private Calculator calc = null!;

        [TestInitialize]
        0 references
        public void Setup()
        {
            calc = new Calculator();
        }

        [TestMethod]
        0 references
        public void Add_ReturnsCorrectSum()
        {
            Assert.AreEqual(7, calc.Add(3, 4));
        }

        [TestMethod]
        0 references
        public void Subtract_ReturnsCorrectDifference()
```

100 % No issues found Ln: 13, Ch: 28 SPC CRLF UTF-8 with BOM

Output

Show output from: Build

```
Build started at 11:20 AM...
========== Build: 0 succeeded, 0 failed, 2 up-to-date, 0 skipped ==========
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```

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Solution Explorer Git Changes

Properties

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File Edit View Git Project Build Debug Test Tools Extensions Window Help Search CalculatorLibrary

Debug Any CPU CalculatorLibrary

CalculatorTests.cs Test1.cs Calculator.cs

CalculatorLibrary.Tests CalculatorLibrary.Tests.CalculatorTests Setup()

```
25          public void Subtract_ReturnsCorrectDifference()
26          {
27              Assert.AreEqual(6, calc.Subtract(10, 4));
28          }
29
30          [TestMethod]
            0 references
31          public void Multiply_ReturnsCorrectProduct()
32          {
33              Assert.AreEqual(12, calc.Multiply(3, 4));
34          }
35
36          [TestMethod]
            0 references
37          public void Divide_ReturnsCorrectQuotient()
38          {
39              Assert.AreEqual(2, calc.Divide(10, 5));
40          }
41
42          [TestMethod]
            0 references
43          public void Divide_ByZero_ShouldThrow()
44          {
45              try
46              {
47                  calc.Divide(10, 0);
48                  Assert.Fail("Expected DivideByZeroException was not thrown.");
49              }
50              catch (DivideByZeroException)
```

100 % No issues found Ln: 13, Ch: 28 SPC CRLF UTF-8 with BOM

Output

Show output from: Build

```
Build started at 11:20 AM...
========== Build: 0 succeeded, 0 failed, 2 up-to-date, 0 skipped ==========
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```

Solution Explorer

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Solution 'CalculatorLibrary' (2 of 2 proj

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Solution Explorer Git Changes

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CalculatorTests.cs Test1.cs Calculator.cs

CalculatorLibrary.Tests CalculatorLibrary.Tests.CalculatorTests Setup()

```
43          public void Divide_ByZero_ShouldThrow()
44          {
45              try
46              {
47                  calc.Divide(10, 0);
48                  Assert.Fail("Expected DivideByZeroException was not thrown.");
49              }
50              catch (DivideByZeroException)
51              {
52
53              }
54          }
55
56          [TestMethod]
            0 refe
57          public
58          {
59              As
60          }
61
62          [TestM
            0 refe
63          public
64          {
65              As
66          }
67      }
68  }
69
```

class Microsoft.VisualStudio.TestTools.UnitTesting.TestMethodAttribute (+ 1 overload)
This attribute is used to mark test methods.

- When using other attributes like DataRowAttribute or DynamicDataAttribute, the use of TestMethodAttribute is still required.
- Test methods must be:
  - public, or if DiscoverInternalsAttribute is used then it can be internal.
  - not static
  - not generic
  - not abstract
  - return type is either void, Task, or ValueTask. If void, then it shouldn't be async.

100 % No issues found Ln: 13, Ch: 28 SPC CRLF UTF-8 with BOM

Output

Show output from: Build

```
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Solution Explorer Git Changes

Properties

Ready Test Explorer Add to Source Control Select Repository

Test Explorer

Search (Ctrl+I)

Test run finished: 8 Tests (8 Passed, 0 Failed, 0 Skipped) run in 139 ms

0 Warnings 0 Errors

| Test | Duration | Traits | Error Message |
| --- | --- | --- | --- |
| CalculatorLibrary.Tests (8) | 307 ms | | |

Run | Debug | Profile

**Group Summary**

CalculatorLibrary.Tests

Tests in group: 8

Total Duration: 307 ms

Outcomes

8 Passed